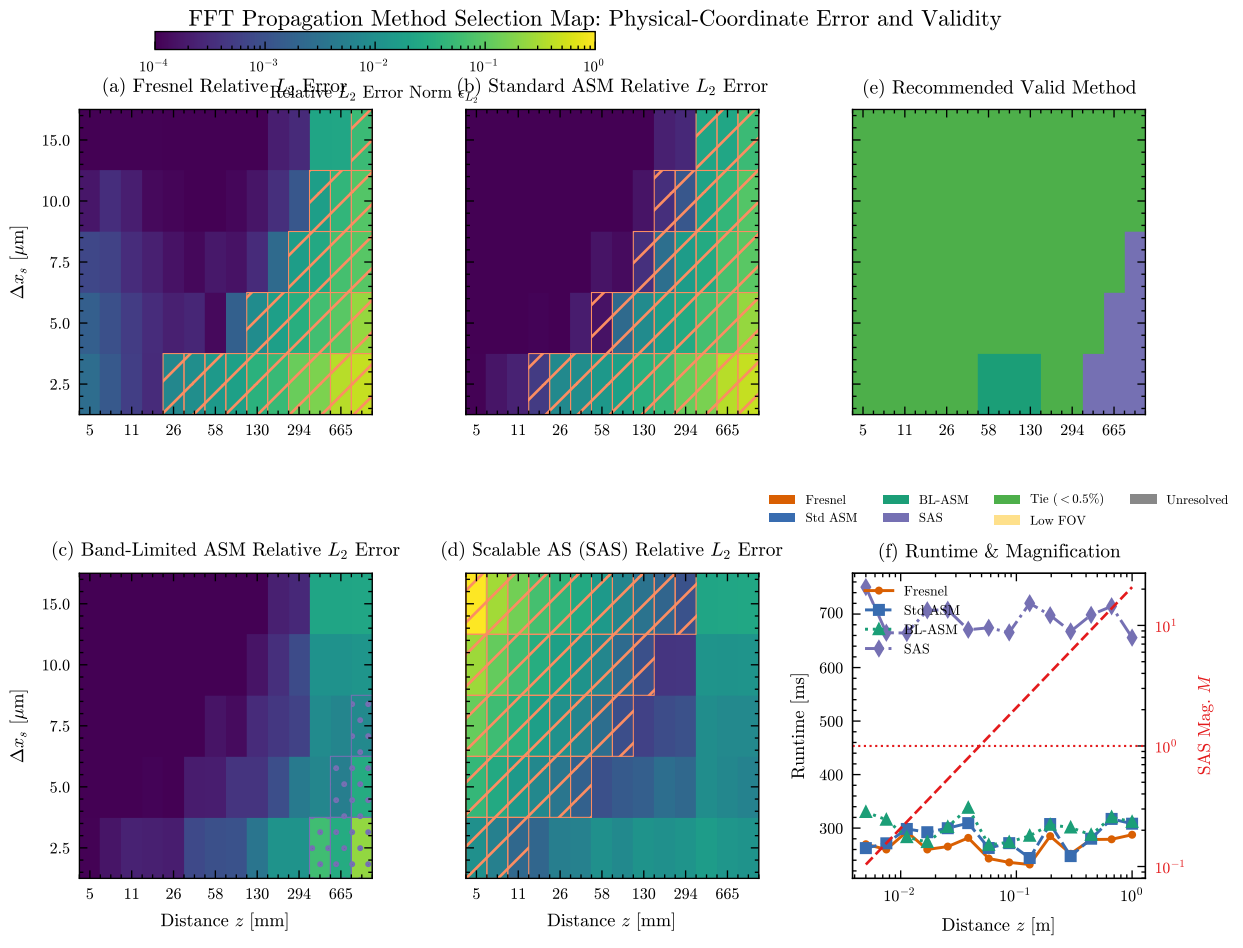

FFT Propagation Method Selection Map: Physical-Coordinate Error and Validity
Relative L_2 Error Norm
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
(a) Fresnel Relative L_2 Error
(b) Standard ASM Relative L_2 Error
(e) Recommended Valid Method
(c) Band-Limited ASM Relative L_2 Error
(d) Scalable AS (SAS) Relative L_2 Error
(f) Runtime & Magnification
Δx_s [μm]
15.0
10.0
7.5
5.0
2.5
5
11
26
58
130
294
665
Distance z [mm]
Fresnel
Std ASM
BL-ASM
SAS
Tie ($<0.5\%$)
Low FOV
Unresolved
Runtime [ms]
300
400
500
600
700
Distance z [m]
10^{-2}
10^{-1}
10^{0}
SAS Mag. M
10^{1}
10^{0}
10^{-1}